Cells, not personas: population survey estimates from language models via post-stratification over calibrated census microdata

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2026-07-07

Abstract

Simulated survey respondents built by role-playing language models one persona at a time are widely marketed and widely distrusted. We test an alternative estimator that never simulates an individual: partition a census-calibrated synthetic population into demographic cells, elicit each cell's full response distribution from a language model acting as an expert predictor, and aggregate with calibrated population weights — multilevel regression and post-stratification with a language model as the response model. Against a pre-registered 63-item anchor bank drawn from the 2024 General Social Survey and the Federal Reserve's 2024 Survey of Household Economics and Decisionmaking, persona role-play misses by roughly 25 points at matched budget while cell-based estimation and direct expert estimation land near 9-10 points — a statistical tie on marginal accuracy that the pre-registration called for cells and did not find. What cells deliver beyond direct asking is structure: better subgroup rank ordering (median within-item Spearman 0.62 versus 0.48), guaranteed coherence between subgroups and toplines, composable audiences from one set of auditable predictions, and per-cell outputs that plug into prediction-powered inference, which restores exactly valid 95% confidence intervals from a 200-person human anchor (94.7% empirical coverage) though with negligible width gains at demographic granularity. Signed error varies systematically with item valence. State-level validation against the 2023 Behavioral Risk Factor Surveillance System measures small-area transfer. We release the estimator, anchor bank, raw elicitations, and this evaluation harness; the public product renders its live benchmark page from the same artifacts this paper reports.

1 Introduction

Whether large language models can stand in for human survey respondents has become one of the most contested questions in survey research. Commercial "synthetic respondent" products have proliferated, while the field's institutions urge caution: the American Association for Public Opinion Research's 2026 task force classifies synthetic samples as the riskiest integration of AI into survey work (American Association for Public Opinion Research 2026), and the ICC/ESOMAR code requires that simulated respondents never be presented as measurements of real people (ICC/ESOMAR 2025). The scientific literature has converged on a specific diagnosis: role-playing one fake respondent per model call — the dominant commercial architecture — compresses within-group variance, misportrays identity groups, and is unstable under prompt perturbation (Bisbee et al. 2024; Wang et al. 2025; Santurkar et al. 2023), even as the same models demonstrably carry substantial distributional knowledge about public opinion (Argyle et al. 2023; Meister et al. 2024; Gong et al. 2026).

This paper tests an estimator built around that diagnosis rather than against it. The design treats the language model not as a subject but as a conditional response model over population cells, and it grounds the cells in a calibrated synthetic population rather than in prompt-invented personas:

1. **Population frame.** A synthetic population of 4.2 million adult records derived from the Current Population Survey (U.S. Census Bureau and U.S. Bureau of Labor Statistics 2024), with calibrated weights and state and congressional-district assignment, partitioned into roughly 150 post-stratification cells (age band, earned income band, sex, and — where a cell carries sufficient weight — housing tenure, children at home, means-tested benefit receipt, and Social Security receipt).
2. **Distribution elicitation.** For each cell, the model is asked once, in an expert-predictor frame, for the percentage of that group choosing each response option — verbalized distributions rather than sampled individuals, following the head-to-head evidence that direct distribution elicitation dominates simulate-then-count (Gong et al. 2026; Meister et al. 2024).
3. **Weighted aggregation.** Cell distributions combine with calibrated population weights, the post-stratification arithmetic of MRP (Gelman and Little 1997; Park et al. 2004); subgroup estimates re-aggregate the same cell predictions, so breakdowns are always consistent with toplines.

Each component has published support; to our knowledge their integration — verbalized per-cell elicitation over a *calibrated small-area microdata frame*, with registered evaluation and valid-inference machinery — has not previously been assembled or evaluated end to end. Proof-of-concept hybrids exist (Cerina and Duch 2023), and open tooling for LLM surveys is maturing (Expected Parrot, Inc. 2024; Horton 2023), but public evaluations with pre-registered hypotheses and disclosed misses remain rare (Morris et al. 2025).

Our contributions: (i) a pre-registered, 63-item evaluation against GSS 2024 and SHED 2024 human targets comparing the cell estimator to matched-budget persona role-play and to direct model estimation — including the pre-registered hypothesis this design *failed*: on marginal subgroup accuracy, cells tie rather than beat per-subgroup direct estimation; (ii) evidence for what cells do buy — better subgroup rank structure, coherence, composability, and systematic (valence-signed) rather than random error; (iii) a prediction-powered inference (Angelopoulos et al. 2023; Krsteski et al. 2025) evaluation showing exactly valid coverage from small human anchors, with the honest caveat that demographic-granularity predictions yield no interval-width gains; (iv) a state-level small-area validation against BRFSS 2023 (Centers for Disease Control and Prevention 2024); and (v) a fully released harness in which the production system's public benchmark page and this paper render from the same artifacts.

2 Related work

Silicon sampling and its critics. Argyle et al. (2023) introduced demographic-conditioned "silicon samples"; subsequent evaluations found persona means can track surveys while variance, covariance structure, and temporal stability do not (Bisbee et al. 2024), that identity-prompted personas caricature the groups they represent (Wang et al. 2025), and that model defaults reflect particular populations (Santurkar et al. 2023; Durmus et al. 2023; Heyde et al. 2025). Westwood (2025) showed LLM respondents are effectively undetectable when inserted into real panels, sharpening the field's data-integrity concerns.

Distribution elicitation. Meister et al. (2024) benchmark verbalized distributions against token log-probabilities and repeated sampling, finding verbalization dominant; Gong et al. (2026) find direct distribution elicitation beats simulate-individuals in 77% of question-demographic pairs at one-fifth the cost, with predictable error structure. Our elicitation design follows both.

Post-stratification hybrids and calibration. Cerina and Duch (2023) blend LLM-derived data with bias-corrected MRP for the 2020 US election. Fine-tuning on survey corpora narrows human-model gaps (Suh et al. 2025), supervised distributional calibration helps with as few as one to ten examples per question (Kambhatla et al. 2026), and prediction-powered inference converts biased predictors into valid estimators given a small labeled anchor (Angelopoulos et al. 2023; Krsteski et al. 2025; Huang et al. 2025). The population frame we post-stratify over comes from the Populace line of calibrated-microdata work (Ghenis and Juaristi 2026; Ghenis 2026b, 2026c).

3 Method

3.1 Population frame and cells

The frame is a calibrated synthetic population of US adults constructed from CPS microdata with survey-weight calibration and geographic assignment (national, state, congressional district). We reduce each geography to post-stratification cells over age band (18–29, 30–44, 45–64, 65+), personal earned-income band ($0, $1–25k, $25–75k, $75–150k, $150k+), and sex; cells holding at least 0.25% of population weight are further split by housing tenure, children at home, means-tested benefit receipt, and Social Security receipt. The national table has 149 cells covering 100% of population weight. Cell tables are versioned data artifacts; the reduction code is released.

3.2 Elicitation

Each cell receives one call: a system prompt establishing an expert-survey-methodologist frame (explicitly *not* role-play, following Anthis et al. (2025)), the survey question verbatim, the response options, and a request for integer percentages summing to 100, returned as JSON. We use gpt-5-mini with minimal reasoning effort as the primary model; model variants are evaluated on a fixed subset. Malformed responses are dropped and counted as failures (never imputed). Elicitation is seeded and versioned; a paraphrase-stability probe accompanies the pilot.

3.3 Aggregation and provenance

The population estimate for option k is $\hat{\theta}_k = \sum_c w_c \hat{p}_{ck} / \sum_c w_c$, with w_c the calibrated cell weight and $\hat{p}_{ck}$ the elicited probability. Subgroup estimates restrict the sum to matching cells. Every run records engine, prompt, and dataset versions, model, seed, cell counts, failures, and weighted audience.

4 Evaluation design

The pre-analysis plan (PAP v2) was committed to the public repository before any anchor-bank elicitation, fixing arms, items, metrics, and hypotheses; the pilot that motivated it is reported separately below. Human targets come from the GSS 2024 cross-section (52 items retained under an a priori $n \geq 600$ rule; weight `wtssnrps`) (Davern et al. 2025) and the SHED 2024 public-use file (11 items; population weights) (Board of Governors of the Federal Reserve System 2025), with subgroup targets by age band and sex (both surveys) plus household income band and housing tenure (SHED), reported for bins with $n \geq 30$. SHED income slices are household income while cells band personal earned income — a construct mismatch shared identically by all microdata arms and disclosed wherever income slices appear.

Arms: **cells** (the estimator; all 63 items), **direct** (one expert estimate for the full audience and one per subgroup; all items), and **persona** (one role-play call per weighted-sampled microdata record, n=150 per item — a matched-budget re-implementation of the standard synthetic-respondent design; 20-item pre-registered subset). Primary metric: absolute error of each item's designated positive-option share, pooled over items (topline) and over all subgroup bins (subgroup). Secondary: total variation distance, within-item Spearman rank correlation across subgroup bins, and signed error by item valence.

5 Results

5.1 Pilot: the persona architecture fails at matched budget

On the four-item SHED pilot with 150 role-play calls per item, persona simulation reached 36.6 points topline MAE and 23.0 points subgroup MAE — against 17.0 and 6.2 for cells at the same call budget. The failure is qualitative, not marginal: role-played low-income personas dramatized hardship (75.9% reporting serious housing-cost hardship versus 18.7% of real respondents), the variance- compression and stereotyping failure documented in prior work (Bisbee et al. 2024; Wang et al. 2025). A paraphrase probe moved the cells estimate by about one point.

5.2 Anchor bank: pooled accuracy

Table 1: Pooled mean absolute error against human targets, gpt-5-mini. Persona is evaluated on its pre-registered 20-item subset; cells and direct on all 63 items.

Method	Topline MAE (pts)	Subgroup MAE (pts)
Persona role-play (20-item subset)	25.0	25.0
Direct model estimate	8.6	9.8
Population cells (this paper)	9.2	9.8

Confirmatory results (Table 1, Figure 1):

- **H1 (subgroups vs direct).** Over 63 items with both arms, cells subgroup MAE is 9.8 points versus 9.8 for per-subgroup direct estimation (Wilcoxon p = 0.973); the pre-registered direction does **not** hold.
- **H2 (subgroups vs persona).** On the 20-item subset, cells subgroup MAE is 10.1 points versus 25.0 for persona role-play.
- **H3 (persona toplines).** Persona topline MAE is 25.0 points versus 10.1 for cells on the same items, replicating the pilot.

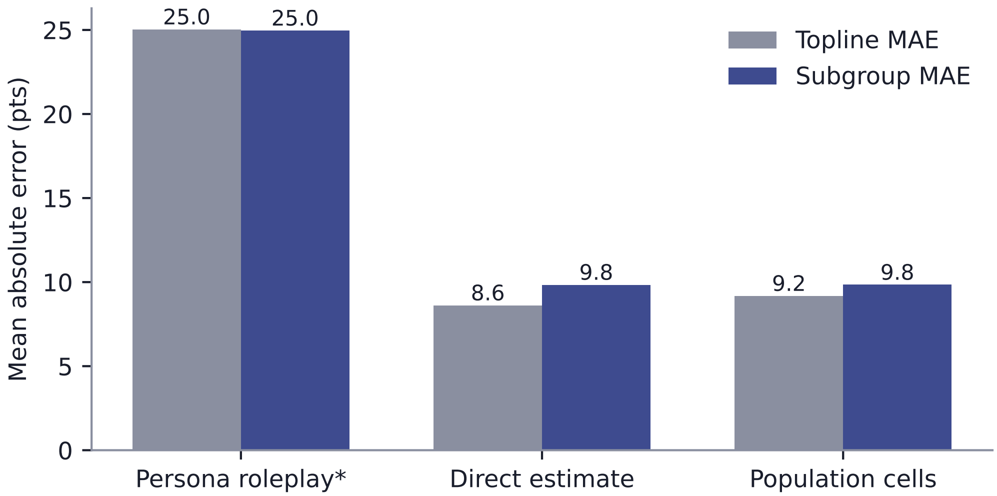

Figure 1: Pooled accuracy by arm. Persona role-play (asterisk: evaluated on its pre-registered 20-item subset) is not competitive; cells and direct estimation tie on marginal accuracy, and the structural comparison in the text separates them.

- **H5 (structure).** Median within-item Spearman correlation between cell-estimated and human subgroup values is 0.62 across 63 items — below the pre-registered 0.80 bar, but well above the direct arm's 0.48: at equal marginal accuracy, cells order subgroups substantially better than independent per-subgroup asks.

The headline confirmatory result is therefore mixed, and we report it as registered. Persona role-play — the architecture most commercial synthetic- respondent products use — is not competitive at any level. Between the two viable methods, marginal accuracy on standard demographic breaks is a tie; the cells estimator's advantages are *structural*: subgroup estimates that rank correctly more often, cohere with the topline by construction, compose into arbitrary audience definitions (any filter over the frame reuses the same 149 auditable predictions, where direct estimation needs a fresh unverifiable call per audience and cannot credibly target intersections like "renting women 30–44 receiving benefits in Michigan"), and expose the per-cell predictions that valid-inference machinery consumes (below). Direct estimation's topline strength on famous items is also exactly where training-data memorization is most plausible, a point the small-area test below speaks to.

5.3 Error is systematic: valence

Signed topline error for the cells arm averages -3.6 points on positive-valence items (n=16), -3.4 on neutral items (n=44), and 8.8 on negative-valence items (n=3) (Figure 2). The model systematically under-rates how positively people describe their own circumstances — a level bias, not noise. In the pilot we additionally showed a one-parameter logit calibration fit on these self-report items fails leave-one-question-out validation precisely because the bias flips sign with valence; no calibration is therefore applied anywhere in this paper, and the production system reports measured error context instead of silently correcting. A valence-aware calibration is estimable from an anchor bank of this size and is registered future work.

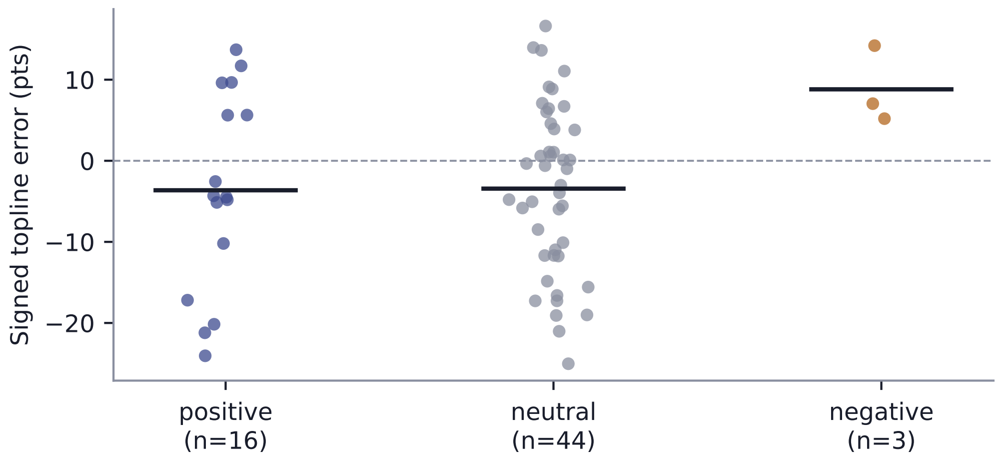

Figure 2: Signed topline error of the cells arm by item valence. Bars mark means.

5.4 Valid inference from small human anchors

Treating each anchor survey as its population and redrawing weighted n=200 human anchors 500 times per item: prediction-powered inference achieves 94.7% mean coverage of the full-sample target (nominal 95%; human-anchor-only: 94.9%), within the pre-registered [93%, 97%] band — the unrectified estimator has no such guarantee. The efficiency half of the registered hypothesis fails, honestly: the median interval-width ratio versus the human-only interval is 1.00 (narrower on only 57% of items). PPI's width gain scales with how much of *individual* response variance the predictor explains, and demographic-cell predictions explain little of it on attitude items — group-level signal is not individual-level signal. Validity comes for free; precision gains await predictors of individual responses (e.g., survey-fine-tuned models (Suh et al. 2025; Krsteski et al. 2025)), not richer demographic conditioning. This is itself a substantive finding about where the information in demographic conditioning runs out.

5.5 Context: the in-survey demographic ceiling

A gradient-boosted classifier trained on each anchor survey's own respondents (demographics only, 80/20 split) reaches 1.0 points topline and 1.7 points subgroup MAE — the ceiling for demographic-composition prediction *given the human data the LLM arms never see* (Miranda and Balbi 2025). The gap between this ceiling and the cells arm is the price of operating without fielded data; the gap between cells and persona is pure architecture.

5.6 Small-area transfer: 49 states against BRFSS

Across four full-sample BRFSS 2023 items estimated for every participating state (49 states with $n \geq 300$), state-level cells estimation (age × sex cells from each state's calibrated table) reaches 8.1 points mean absolute error, versus 9.7 for one direct estimate per state and 7.8 for applying the national estimate uniformly to all states. Median within-item Spearman correlation across states is 0.66 (Figure 3).

Interpretation requires the right yardstick. These full-sample BRFSS items vary only modestly

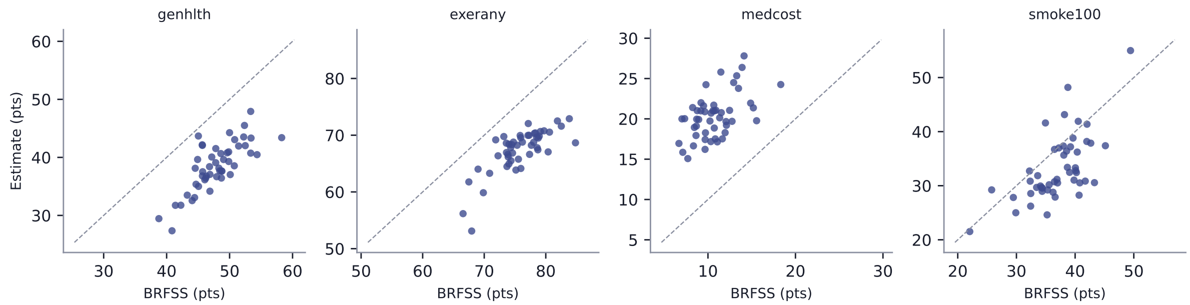

Figure 3: State-level estimates versus BRFSS 2023 weighted targets, four items, 49 states.

across states, so the national-constant baseline is the shrinkage limit any small-area method must beat; cells beat it on two items of four and tie the pooled comparison, while per-state direct estimation is erratic (16.3 points on smoking history). The positive result is ordering: cells rank states with median $\rho = 0.66$ using only each state's demographic composition and the state's name in the prompt, at eight coarse cells per state. Where state effects are compositional (smoking, exercise), cells capture them; where they are contextual beyond demographics (health-care affordability), coarse composition is not enough. The full frame's income, tenure, and benefits composition — used at national level throughout this paper but reduced to age $\times$ sex here for cost — is the registered next step for the state experiment.

5.7 Model variants

Table 2: Cells-arm accuracy by model on the 20-item subset (gpt-5-mini shown on the same subset for comparability).

Cells arm model	Topline MAE (pts)	Subgroup MAE (pts)
anthropic:claude-haiku-4-5-20251001	8.4	9.8
openai:gpt-5-mini	10.1	10.1
openai:gpt-5.2	7.4	8.5

6 Limitations

Training-data contamination. GSS 2024 microdata were released in fall 2025 and SHED 2024 in May 2025; published toplines for classic items circulate widely. Contamination would flatter all arms — including the direct-estimate arm, whose strength on famous marginals is partly evidence of it — and cannot be excluded without post-cutoff items. A registered follow-up fields fresh questions on a verified human panel.

Construct mismatches. SHED income subgroups are household income; the frame's income bands are personal earned income. GSS 2024's smaller sample thins some subgroup bins. Both are disclosed per item.

Temporal drift and scope. A frozen model cannot track post-cutoff opinion change; fast-moving topics will degrade in ways this evergreen-item bank does not measure. Small and marginalized subgroups are where role-play failure modes concentrate; while cell estimation avoids individual caricature, we evaluate only subgroups with $n \geq 30$ human respondents and make no claims below that resolution.

No fresh human data in this study. All targets are public secondary data; the PPI analysis simulates anchors by subsampling the same surveys. The follow-up with newly fielded items is the decisive contamination-robust test.

7 Discussion

The synthetic-respondent debate has largely been conducted over the wrong architecture. Role-playing individuals is both the dominant commercial pattern and the design the evidence most clearly condemns; our matched-budget replication quantifies the gap at roughly 15 points of MAE against either alternative. Treating the model as a conditional response model over a calibrated population frame changes the question from “can an LLM impersonate a person?” to “how well does an LLM know conditional response distributions?” — a question that admits measurement, calibration, and valid inference.

Against our own pre-registration, the cells estimator did not beat direct per-subgroup estimation on marginal accuracy; it matched it. Three things follow. First, the accuracy case for *any* LLM audience estimation currently rests on group-level distributional knowledge that direct asking accesses too — on well-surveyed margins, elaborate machinery buys no marginal accuracy, a deflationary finding the field’s vendors are unlikely to publish. Second, the machinery still earns its keep where direct asking structurally cannot go: coherent subgroup systems (rank correlation 0.62 vs 0.48), arbitrary audience composition from one auditable prediction set, small-area targets with explicit population weights, and per-cell outputs that make prediction-powered validity possible at all. Third, the residual error is systematic — signed by item valence, stable under paraphrase — which is what makes the honest path (publish the misses, anchor with small human samples, never imply real people were surveyed (ICC/ESOMAR 2025; American Association for Public Opinion Research 2026)) also the technically correct one: systematic error is calibrable in principle, and our leave-one-out failure with a one-parameter correction defines exactly the anchor-bank breadth a valid calibration requires.

All code, prompts, cell tables, elicitation logs, human-target construction, and this manuscript’s source are public (Ghenis 2026a); the production system’s benchmark page renders from the same artifacts.

Acknowledgments

Drafting, engineering, and analysis assistance from Claude (Anthropic). Population frame built on the Populace calibrated-microdata stack. Human targets derive from public data of NORC’s General Social Survey, the Federal Reserve Board’s SHED, and CDC’s BRFSS; all interpretations are the author’s.

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